

# Design and Layout Improvement to GRNsight v2.0: a Web Application and Service for Visualizing Small- to Medium-Scale Gene Regulatory Networks

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LMU Undergraduate Research Symposium

March 25, 2017

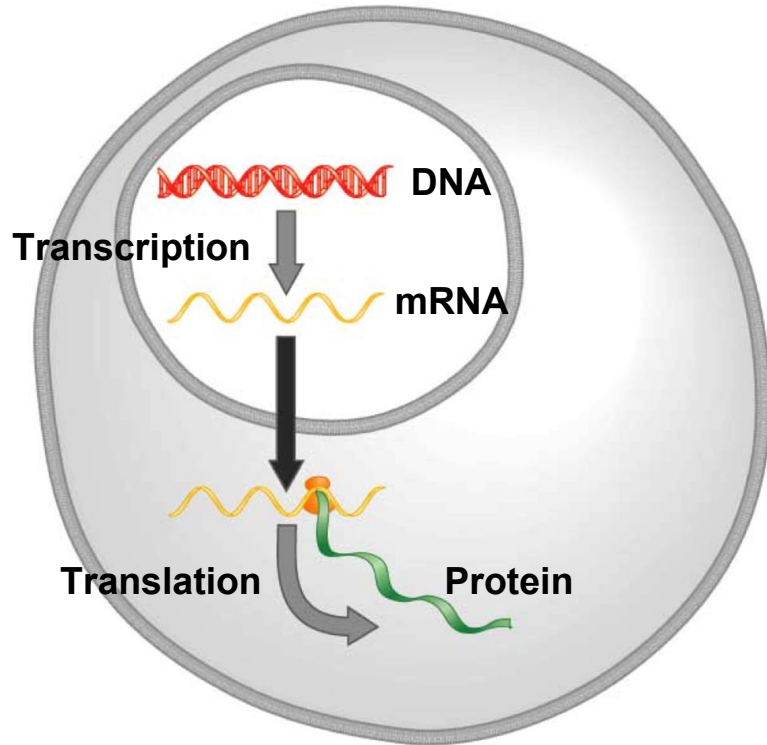
# Outline

- **Gene regulatory networks (GRNs) consist of genes, transcription factors, and the regulatory relationships between them.**
- **GRNsight is an open source web application and service for visualizing models of gene regulatory networks.**
- **Version 1.0 established a testing framework and created an errors and warnings library.**
- **Version 2.0 is in active development, available in beta website, and will allow for greater customization of the graph visuals.**

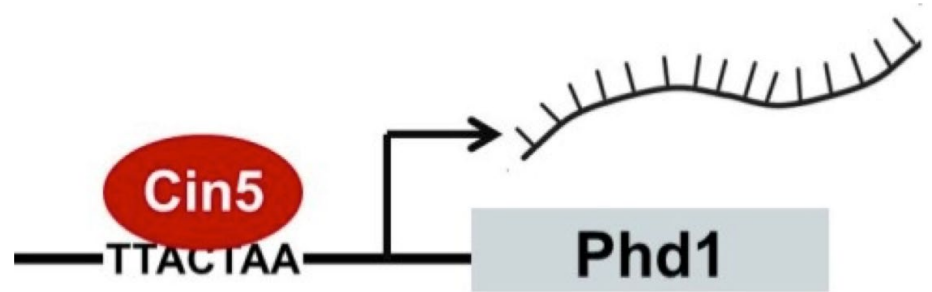
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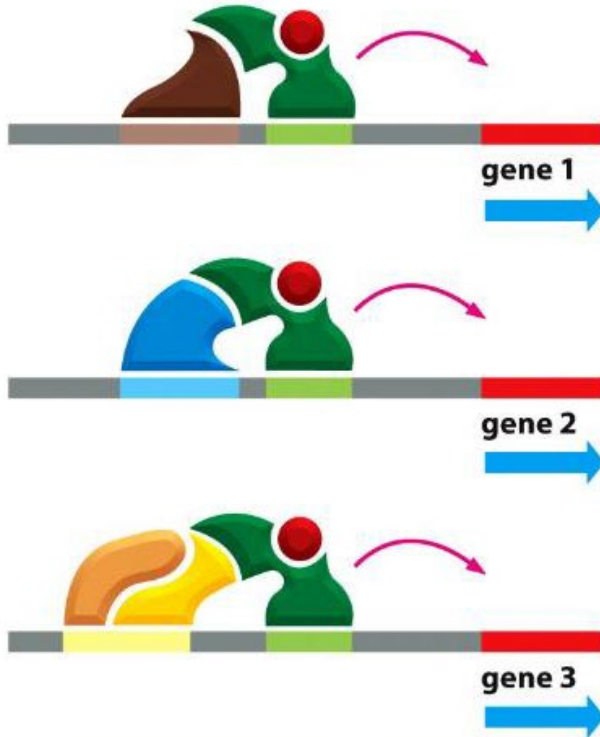
# Transcription Factors Control Gene Expression By Binding to Regulatory DNA Sequences



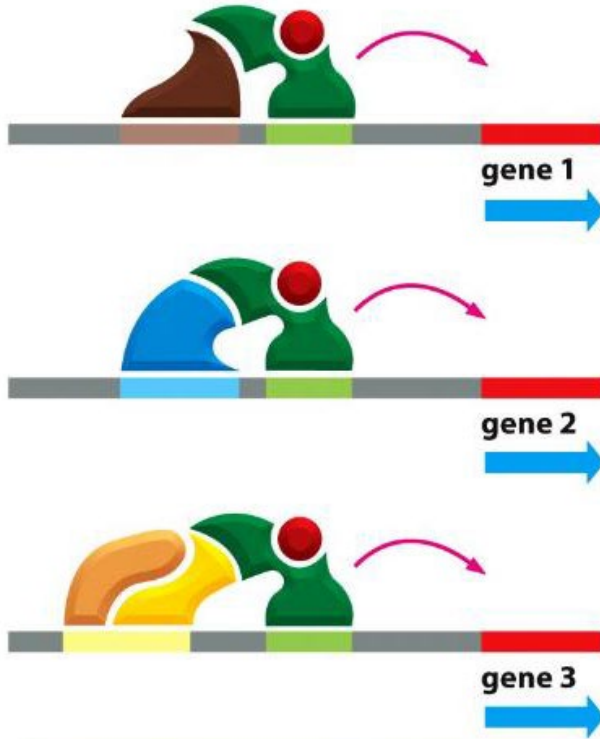
- Activators increase gene expression
- Repressors decrease gene expression
- Gene regulatory networks govern the level of expression of mRNA and proteins from those genes.



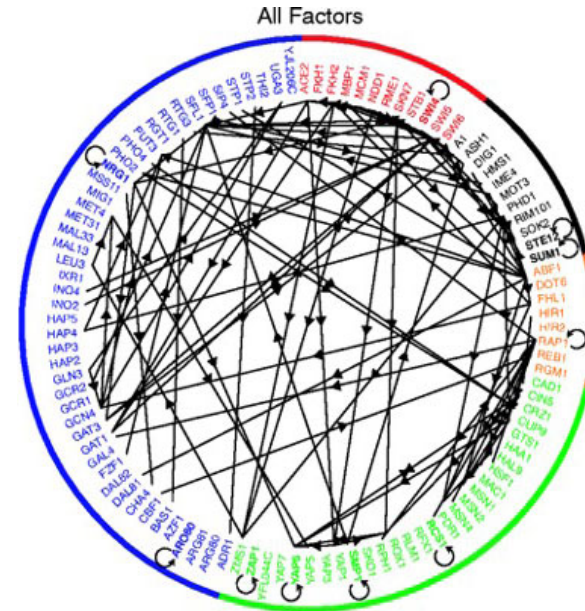
# In Yeast, ~250 Transcription Factors Regulate ~6000 Genes through Combinatorial Control



# In Yeast, ~250 Transcription Factors Regulate ~6000 Genes through Combinatorial Control



Gene Regulatory Network (GRN)  
of 106 Transcription Factors



# GRNsight is Optimized to Visualize Small- to Medium-Scale Networks

| Cytoscape and Gephi                                    | GRNsight                                              |
|--------------------------------------------------------|-------------------------------------------------------|
| Optimized for large-scale graphs.                      | Optimized for small- to medium-scale graphs.          |
| Must be installed.                                     | Exists as a web application.                          |
| Several complex features in addition to visualization. | Easy to use, and easily interpretable visualizations. |

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# GRNsight Accepts Excel Spreadsheets with an Adjacency Matrix

- Regulators and targets are in columns and rows respectively
- Unweighted networks have 1's and 0's, weighted networks have real numbers instead of 1's
- Positive numbers indicate activation and negative numbers indicate repression

cols regulators/rows targets

|    | A                            | B    | C    | D    | E    | F    | G    | H    | I    | J    | K    | L    | M    | N    | O    | P    | Q    | R    | S    | T    | U    | V    |
|----|------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 1  | cols regulators/rows targets | ABF1 | ACE2 | AFT1 | CIN5 | CUP9 | FHL1 | GTS1 | HAL9 | HSF1 | MAC1 | MSN1 | MSN4 | NRG1 | PHD1 | RAP1 | REB1 | ROX1 | RPH1 | SKN7 | YAP1 | YAP6 |
| 2  | ABF1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 3  | ACE2                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 4  | AFT1                         | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    |
| 5  | CIN5                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    |
| 6  | CUP9                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    |
| 7  | FHL1                         | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 8  | GTS1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    |
| 9  | HAL9                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 10 | HSF1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    |
| 11 | MAC1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 12 | MSN1                         | 1    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 13 | MSN4                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 1    | 1    | 0    | 0    | 0    | 0    | 0    |
| 14 | NRG1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 1    | 0    |
| 15 | PHD1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 16 | RAP1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    |
| 17 | REB1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 18 | ROX1                         | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 1    |
| 19 | RPH1                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 20 | SKN7                         | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| 21 | YAP1                         | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    |
| 22 | YAP6                         | 0    | 0    | 0    | 1    | 1    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 1    | 0    | 0    | 1    | 1    |

network

unweighted

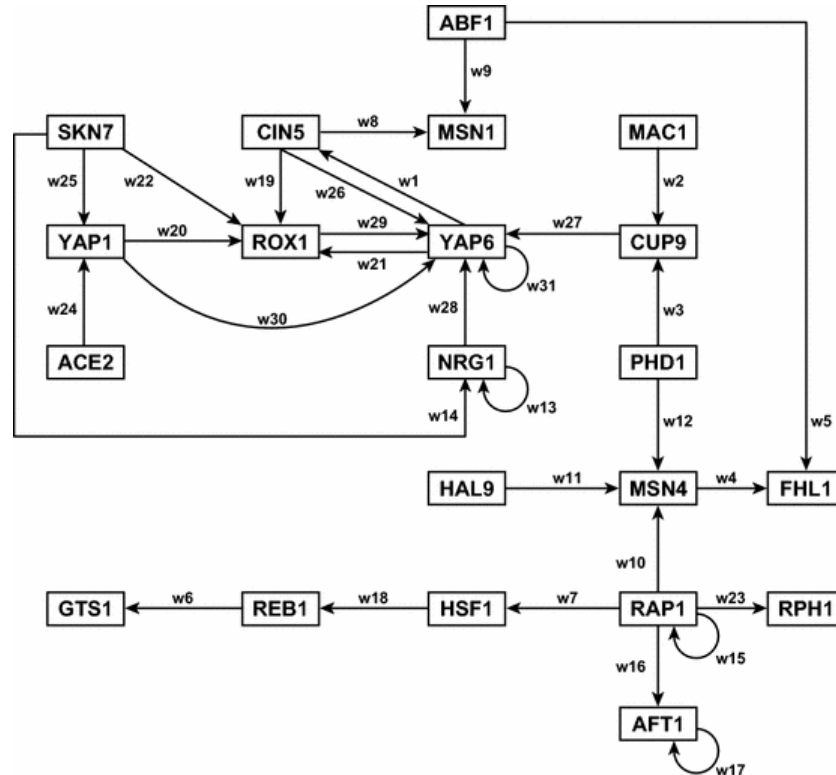
cols regulators/rows targets

|    | A                            | B       | C       | D       | E       | F       | G    | H    | I      | J       | K       | L    | M      | N      | O       | P       | Q      | R       | S       | T      | U       | V       |
|----|------------------------------|---------|---------|---------|---------|---------|------|------|--------|---------|---------|------|--------|--------|---------|---------|--------|---------|---------|--------|---------|---------|
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| 2  | ABF1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 3  | ACE2                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 4  | AFT1                         | 0       | 0       | -0.8966 | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | -0.4030 | 0      | 0       | 0       | 0      | 0       | 0       |
| 5  | CIN5                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | -0.0450 |
| 6  | CUP9                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | -0.1882 | 0    | 0      | 0      | -0.6510 | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 7  | FHL1                         | 0.1562  | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0.6121 | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 8  | GTS1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0.0778  | 0       | 0      | 0       | 0       |
| 9  | HAL9                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 10 | HSF1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | -1.2321 | 0      | 0       | 0       | 0      | 0       | 0       |
| 11 | MAC1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 12 | MSN1                         | -2.9707 | 0       | 0       | 0.9393  | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 13 | MSN4                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 1.4283 | 0       | 0       | 0    | 0      | 0      | 0       | 0.5447  | 1.0131 | 0       | 0       | 0      | 0       | 0       |
| 14 | NRG1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 1.2341 | 0       | 0       | 0      | 0       | -0.1852 | 0      | 0       | 0       |
| 15 | PHD1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 16 | RAP1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | -0.8890 | 0      | 0       | 0       | 0      | 0       | 0       |
| 17 | REB1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | -0.0102 | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 18 | ROX1                         | 0       | 0       | -0.9278 | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0.5744 | -0.4315 | -0.5071 |
| 19 | RPH1                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 1.4999  | 0      | 0       | 0       | 0      | 0       | 0       |
| 20 | SKN7                         | 0       | 0       | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | 0       | 0       |
| 21 | YAP1                         | 0       | -1.3615 | 0       | 0       | 0       | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0      | 0       | 0       | 0      | 0       | 0       | 0      | -0.4082 | 0       |
| 22 | YAP6                         | 0       | 0       | 0       | -0.5312 | -0.1293 | 0    | 0    | 0      | 0       | 0       | 0    | 0      | 0.6215 | 0       | 0       | 0      | -0.7503 | 0       | 0      | 0.0146  | -0.3027 |

network\_optimized\_weights

weighted

# This GRN Diagram Took Several Hours to Create by Hand in Adobe Illustrator



# GRNsight Rapidly Generates GRN Graphs Using Our Customizations to the Open Source D3 Library



## GRNsight

Web app and service for visualizing models of gene regulatory networks.

A Joint Project of the  
LMU Bioinformatics and  
Biomechanics Groups

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Last modified: 8/11/16  
1:55 PM

GRNsight v1.18.1

Welcome to GRNsight. Please select a file to upload.

File Edit Format Help Demo 21-genes\_31-edges\_Schade-data\_input.xlsx

21 nodes  
31 edges

Link Distance (1 - 1000): 500



Charge (-2000 - 0): -1000



Charge Distance (0 - 2000): 1000



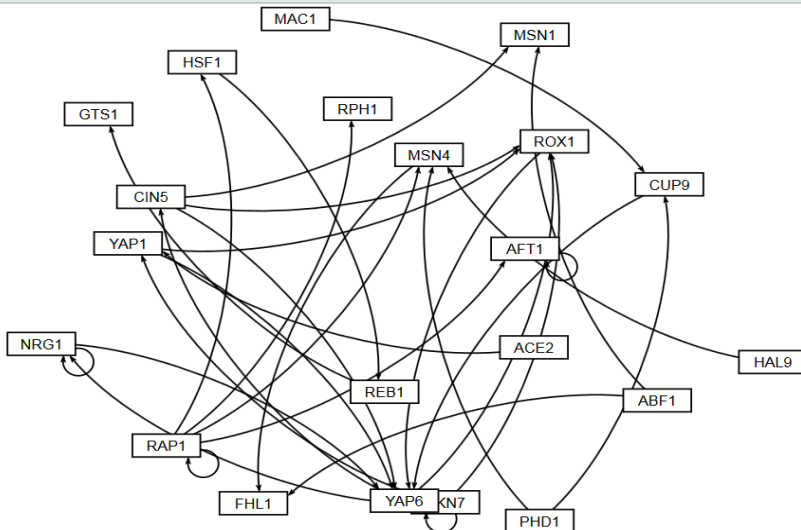
Gravity (0 - 1): 0.10



☐ Lock Force Graph Parameters

Reset Force Graph Parameters

Undo Reset



**10 milliseconds  
to generate**

# Users Can Rearrange Networks Using Drag and Drop



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GRNsight v1.8.1

Welcome to GRNsight. Please select a file to upload.

File Edit Format Help Demo 21-genes\_31-edges\_Schade-data\_input.xlsx

21 nodes  
31 edges

Link Distance (1 - 1000): 500

Charge (-2000 - 0): -1000

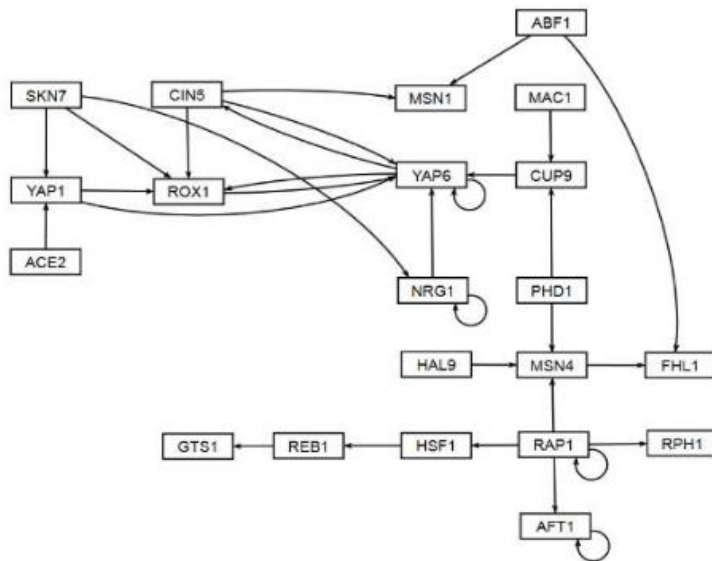
Charge Distance (0 - 2000): 1000

Gravity (0 - 1): 0.10

☐ Lock Force Graph Parameters

Reset Force Graph Parameters

Undo Reset



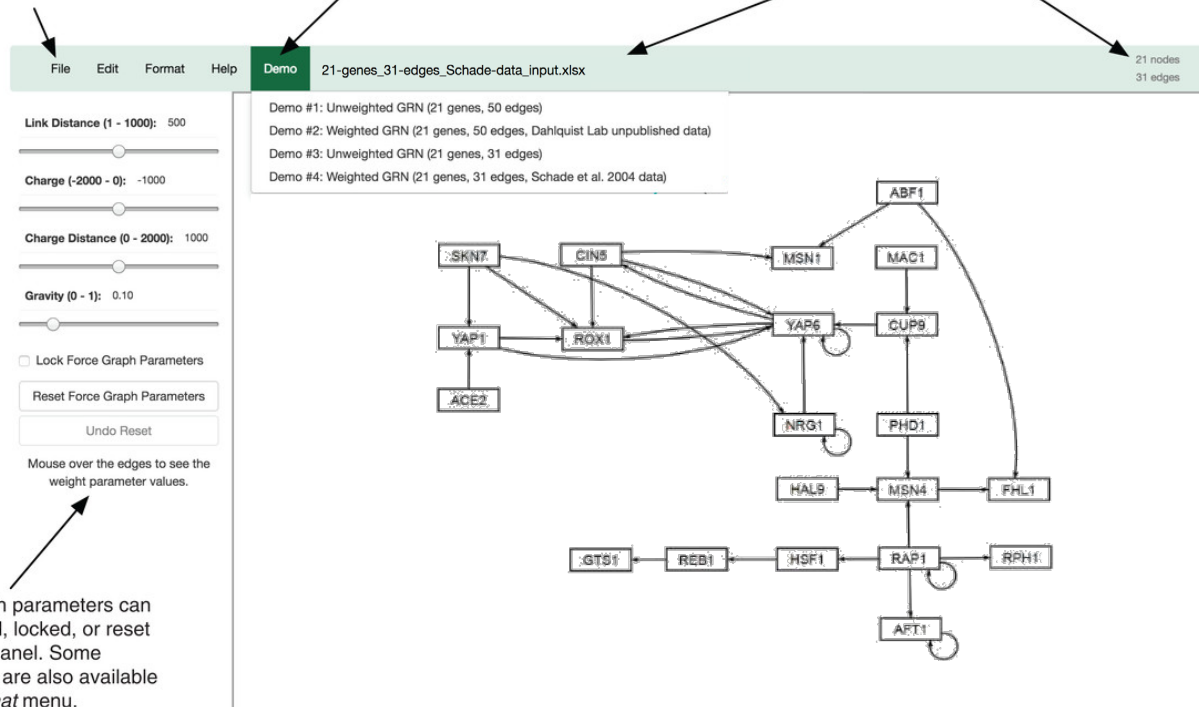
**10 milliseconds to generate,  
5 minutes to arrange**

# GRNsight Automatically Lays Out Weighted and Unweighted Networks

The File menu includes commands for uploading an adjacency matrix in Microsoft Excel (.xlsx) and other formats.

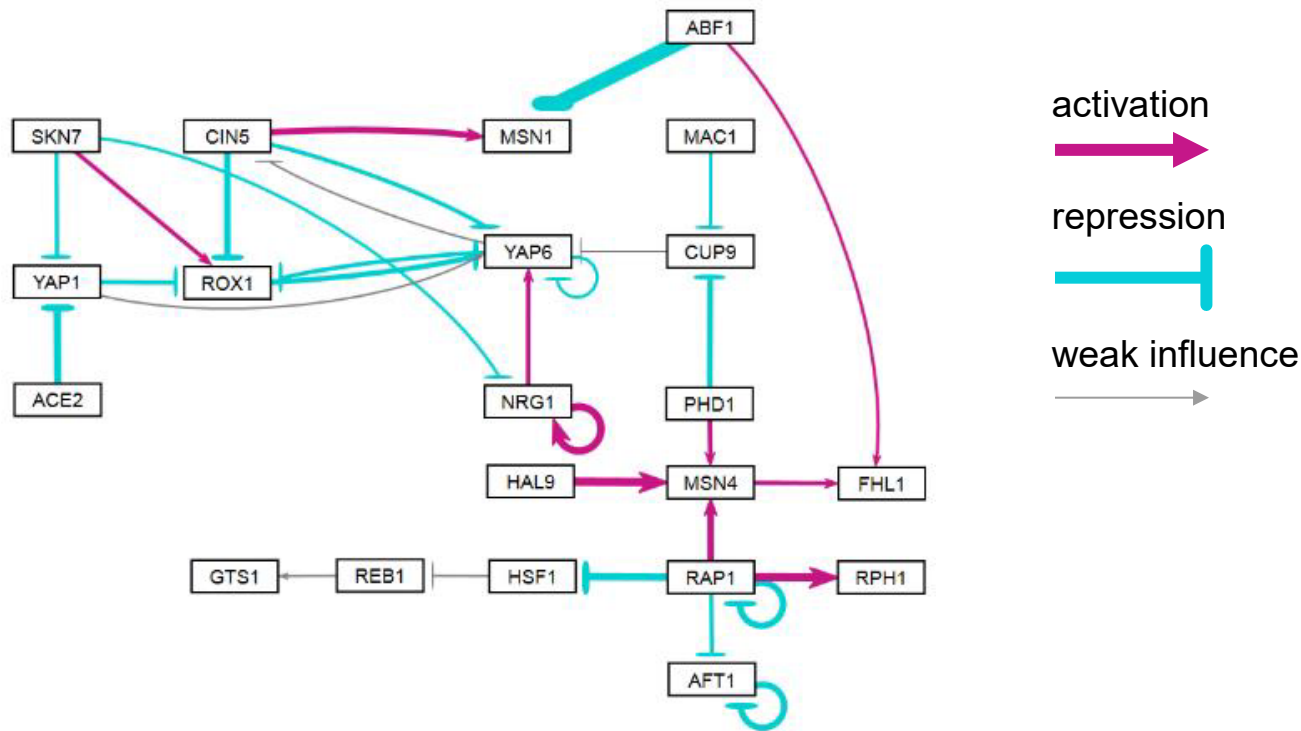
The Demo menu lists four GRNs that have been preloaded into the server.

The status display shows the current filename along with node and edge counts.



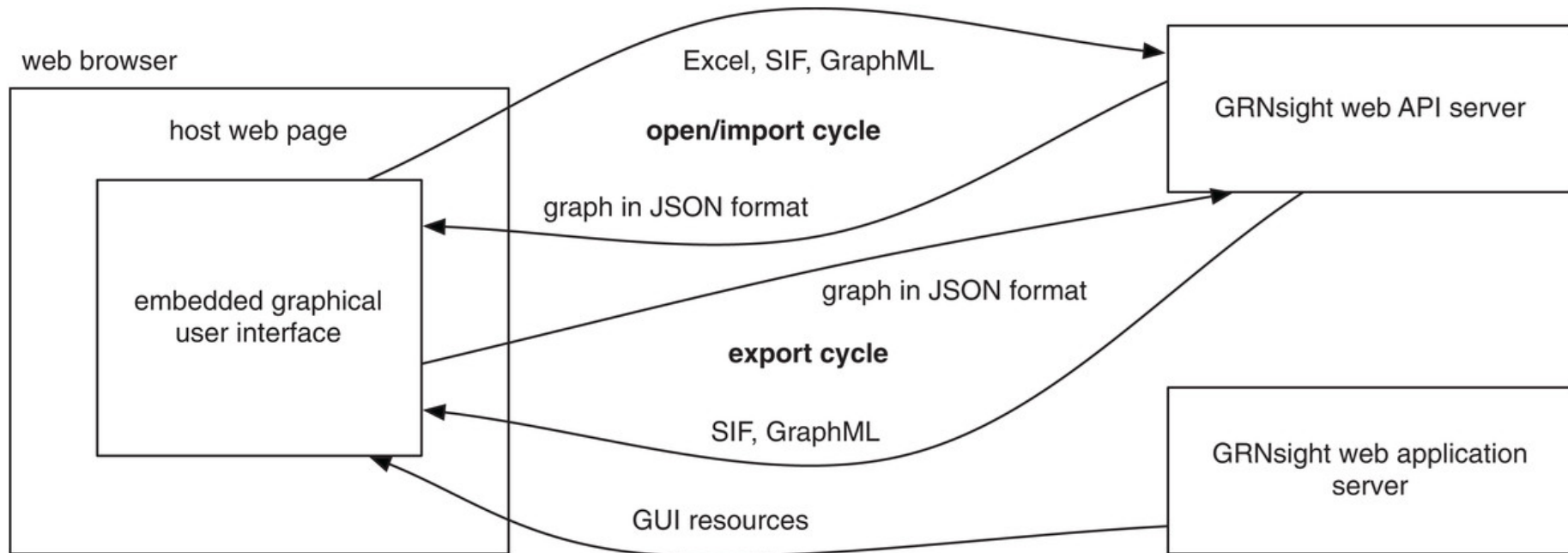
Force graph parameters can be adjusted, locked, or reset using this panel. Some commands are also available in the *Format* menu.

# GRNsight Automatically Lays Out Weighted and Unweighted Networks



GRNsight: colored and variable thickness edges reveal patterns in data

# GRNsight Has a Service-Oriented Architecture



# Outline

- Gene regulatory networks (GRNs) consist of genes, transcription factors, and the regulatory relationships between them.
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- Version 2.0 is in active development, available in beta website, and will allow for greater customization of the graph visuals.

# The Errors and Warnings Library Was Established In v1.0

- When a user uploads a spreadsheet that is formatted incorrectly – GRNsight returns an error or a warning
- Fatal = graph fails to load
- Non-fatal = graph will load
- Warning box can be closed and reopened at any time via a hyperlink underneath Force Graph Parameter sliders

## Error

Your graph failed to load.

The value under gene name ACE2\* contains invalid character. Please ensure all values in the data does not contain special characters except for '-' and '\_'.

Close

Select New File

## Warning

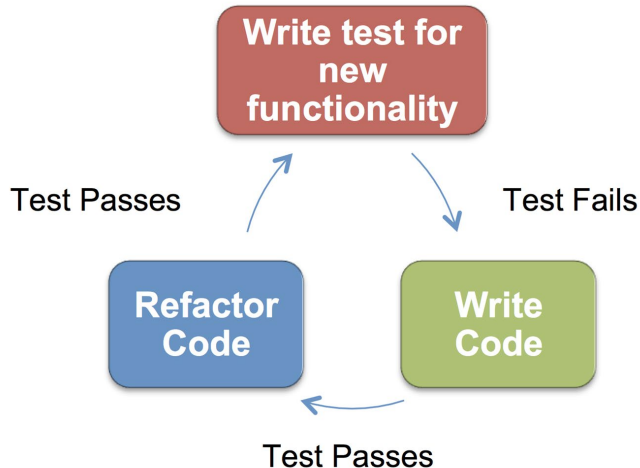
There were 1 warning(s) detected in this file. It is possible that these warnings are the result of extraneous data outside of the matrix, but we recommend you review your file and ensure that it is formatted correctly. The graph will be loaded, but may not be displayed accurately. To view the details of the warning(s), please click on the "Warnings List" below.

[Warnings List](#)

Close

# Test Driven Development (TDD) Ensures GRNsight Works as Expected

- Consists of 230 passing tests covering over 500 test files in beta
- Executed through Mocha, a JavaScript test framework running on node.js and written in Chai, an assertion library for node.js



| Aspect of the code | Test coverage on v1.18.1 (%) | Test coverage on v2.0.1 (%) |
|--------------------|------------------------------|-----------------------------|
| Statements         | 272/371 (73%)                | 344/489 (71%)               |
| Branches           | 158/185 (85%)                | 169/203 (83%)               |
| Functions          | 49/72 (68%)                  | 61/96 (63%)                 |
| Lines              | 272/371 (73%)                | 344/480 (71%)               |

# Outline

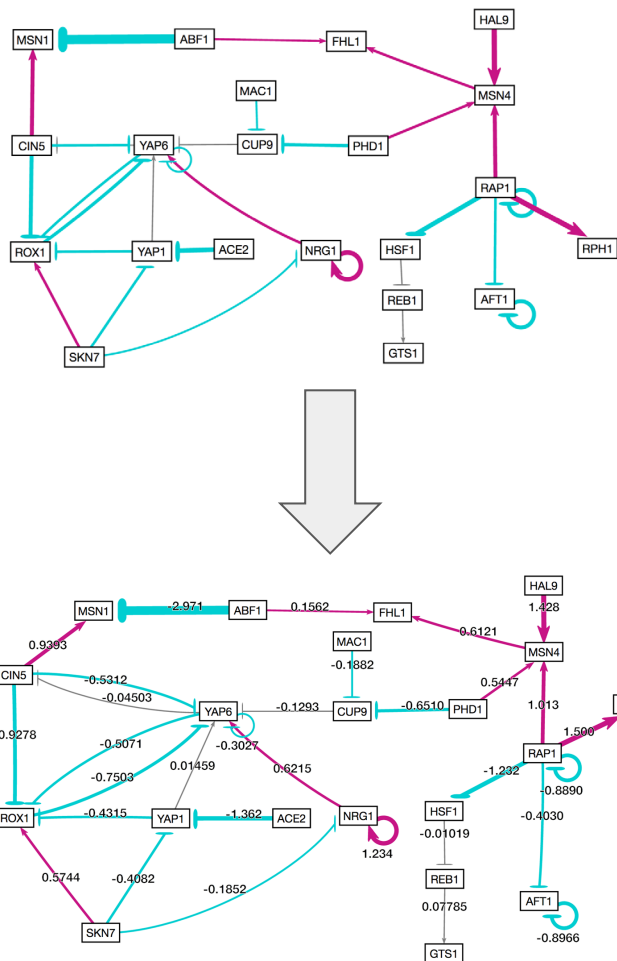
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## Visualization of Edge Weights is User Customizable

- **By default, weights are shown only upon mouse-over of the edge.**
- **Users can choose to always show or always hide weights.**
- **Option exists in sidebar menu, and under Format tab in the File menu.**

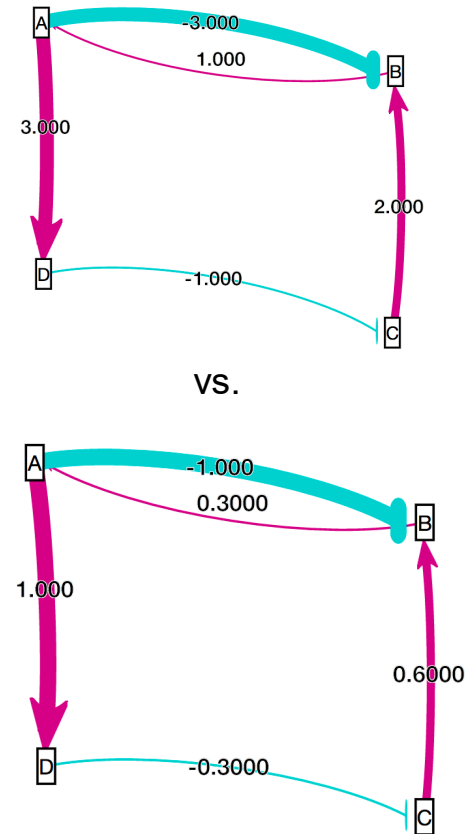
## Hide/Show Edge Weights

- ☒ Show With Mouse Over
- ☐ Always Show Edge Weights
- ☐ Never Show Edge Weights



# A Customizable Normalization Factor Allows Users to Compare Graphs

- Relationship data is normalized to fit within 12 distinct edge thicknesses
- Data from each graph is normalized according to matrix data
- In order to compare graph visualizations, normalization factor must be the same
- User inputs smallest value and largest value in all comparable matrices, and edge thicknesses adjust accordingly



# The viewport was enlarged to allow for easier visualization on different screen sizes

- Three different viewport (visible area) sizes were added to adjust for different screen sizes.
  - Small is best for 13-15" laptops
  - Medium is best for 24" screens
  - Large is best for 32"+ screens
- The initial size of the viewport is chosen automatically based on the amount of available space on the user's screen.
  - It can also be changed via toggle switch.

Toggle Switches

☐ Small

☒ Medium

☐ Large

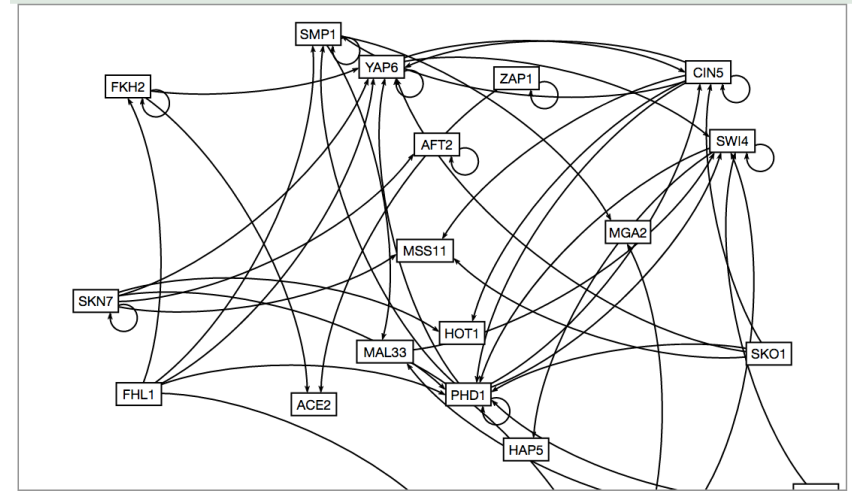
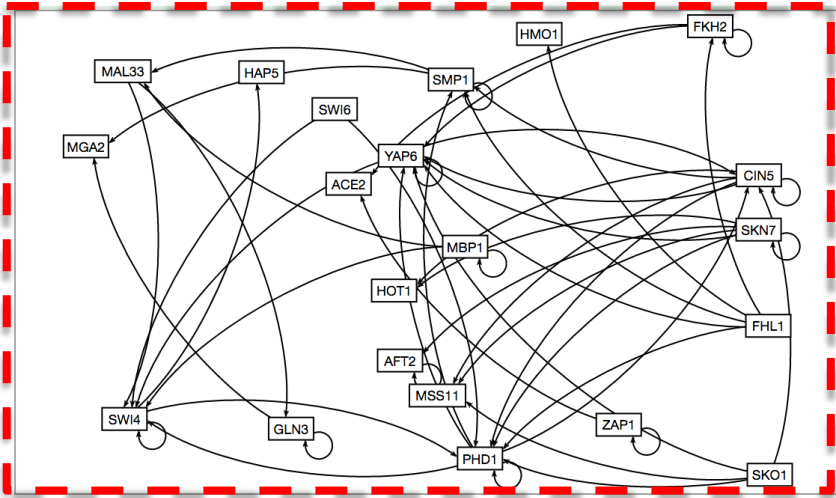
# Viewport automatically detected as small on small screen



# The Drafting Board Was Enlarged for Optimal Graph Loading

- The drafting board (drawable area) was initially too small to allow for the graph to fully expand to an optimal layout.
- Larger viewport sizings allow for the graph to fully expand on load.
- An automatic drafting board resizing algorithm is in development to allow graphs of all sizes to come to an optimal layout.
  - The drafting board will expand past the size of the viewport, and can be fully seen with zooming and scrolling.
- The option of a hard (fixed to viewport) or adaptive (dynamic resizing) drafting board can be selected.

# Adaptive drafting board algorithm extends drafting board beyond viewport



**Zooming and Scrolling Enabled**

# Summary

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# Acknowledgments

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Dr. Kam D. Dahlquist  
Dr. John David N. Dionisio  
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Jen Shin  
Eileen Choe  
Edward Bachoura

GRNmap Team  
Trixie Anne Roque  
Eddie Azinge  
Natalie Williams  
Kristen Horstmann  
Maggie O'Neil  
Brandon Klein  
Justin Kyle Torres

<http://dondi.github.io/GRNsight/>

